Annex III. Debt Sustainability Analysis

Under the baseline projection, Ecuador's public debt is on a sustainable path, thanks to the envisaged fiscal consolidation. Debt is expected to peak in 2020 at around 50 percent of GDP and decline to below 40 percent of GDP by 2024. Gross financing needs are estimated at 8.1 percent of GDP in 2019 but are expected to decline to 5.2 percent of GDP in 2020 and further to 2.7 percent of GDP in the medium term. The debt profile is particularly vulnerable to unexpected and large terms of trade shocks and a sharp deceleration in growth though debt remains sustainable even under the stress scenarios.

Definition and debt profile. Public-sector debt includes the obligations of the non-financial public sector (central government and non-financial sector state-owned enterprises), as reported by the authorities, as well as liabilities under oil related financing, treasury certificates, central bank lending to the government, and other liabilities. Under this measure, public debt more than doubled between 2012 and 2018 (from 17.5 to 46.1 percent of GDP).

Macroeconomic and fiscal assumptions. Growth is projected at about 2.5 percent and inflation (GDP deflator) at about 1 percent over the medium term. The fiscal position is projected to improve to reach a primary surplus of about 5 percent of GDP by 2021 and stay broadly at that level over the medium term. Over the medium-term, the interest rate for private external borrowing is forecasted to converge to about 8 percent from the average of 9,4 in 2019 as a result of the full implementation of the proposed fiscal consolidation. Under these assumptions, the gross financing needs of the public sector are projected to fall from about 8.5 percent in 2018 to 5.2 percent of GDP in 2020 and to 2.7 percent of GDP in the medium term. The public-debt-to GDP ratio is projected to peak in 2020 at 50.4 percent of GDP, and slowly decline to 38.6 percent by 2024.

Stress tests. The fiscal consolidation path would reduce gross financing needs risks significantly in the short- and medium-term. As a result, gross financing needs do not breach the low-risk threshold of 15 percent of GDP in the baseline scenario nor under any of the stress-tests scenarios. The gradual phasing-out of treasury certificates (CETES) over the medium term would also reduce gross financing needs a source of risks going forward. The public-debt-to GDP ratio is below the critical threshold for emerging economies of 70 percent of GDP in the baseline and in the stress-tests scenarios but is particularly sensitive to a real GDP growth shock, an oil shock or a combined shock. In case of a real GDP growth shock, debt would reach 55.8 percent of GDP in 2021, while in the case of a combined macro-fiscal shock debt would increase to 56.4 percent of GDP in the same year. In both cases, debt is expected to gradually decline over the medium-term. A terms of trade shock -like the oil shock experienced in 2014−15 - could pose concerns to sustainability as debt would reach almost 62.5 percent of GDP at the end of 2022. However, as in the previous cases, it would gradually decline reaching 58.8 percent in 2024. Debt is expected to decline gradually and not differ significantly from the medium-term level in the baseline scenario even in the event that spreads will remain at the 2019 average level for a prolonged period of time. This is owning to the fact that the next bond issuance at market rates is envisioned only in 2022, hence, the increase in interest payments will be relatively modest.

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Risks and vulnerabilities. The public DSA risk assessment identifies the ratio of public debt held by non-residents as a high risk for Ecuador, while the recent widening of the spread has increased once-more the market perception of risk to high. At about 73 percent of the total debt, non-residents' holdings of Ecuador's public debt are high but about half of external debt is owed to official creditors and therefore mitigates this vulnerability. Thanks to the recent liability management operation, near-term external debt rollover risks are limited.

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Ecuador Public Sector Debt Sustainability Analysis (DSA) - Baseline Scenario

(in percent of GDP unless otherwise indicated)

Debt, Economic and Market Indicators [1/]

	Actual			Projections					
	2008-2016 [2/]	2017	2018	2019	2020	2021	2022	2023	2024
Nominal gross public debt	24.3	44.6	46.1	49.6	50.1	48.1	44.8	41.5	38.2
Public gross financing needs	7.5	16.7	8.5	8.1	5.2	3.4	4.1	3.8	3.3
Net public debt	19.0	39.6	40.3	42.2	41.5	38.8	35.9	32.9	30.0
Real GDP growth (in percent)	3.5	2.4	1.3	-0.5	0.2	1.6	2.7	2.5	2.5
Inflation (GDP deflator, in percent)	4.2	1.9	1.8	0.2	0.5	0.7	1.0	1.1	1.2
Nominal GDP growth (in percent)	8.0	4.4	3.1	-0.3	0.7	2.4	3.7	3.6	3.7
Effective interest rate (in percent) [4/]	4.6	5.1	5.8	5.6	6.0	5.9	5.7	5.7	6.3

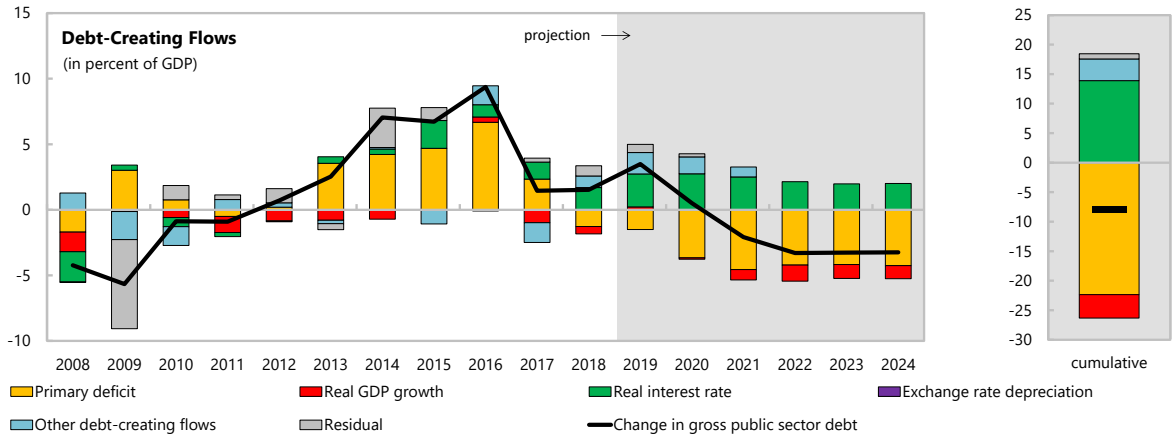
As of December 11, 2019

Sovereign Spreads	
EMBIG (bp) 3/	972
5Y CDS (bp)	n.a

Ratings	Foreign	Local
Moody's	B3	B3
S&Ps	B-	B-
Fitch	B-	B-

Contribution to Changes in Public Debt

	Actual			Projections							
	2008-2016	2017	2018	2019	2020	2021	2022	2023	2024	cumulative	debt-stabilizing primary balance [9/]
Change in gross public sector debt	1.6	1.5	1.5	3.5	0.5	-2.1	-3.3	-3.3	-3.2	-7.9	
Identified debt-creating flows	1.7	1.1	0.7	2.9	0.3	-2.1	-3.3	-3.3	-3.2	-8.8	
Primary deficit	2.3	2.3	-1.3	-1.5	-3.7	-4.6	-4.2	-4.2	-4.3	-22.4	1.0
Primary (noninterest) revenue and grants	35.4	32.0	36.1	34.2	36.2	35.5	34.9	34.9	34.9	210.6	
Primary (noninterest) expenditure	37.7	34.4	34.9	32.7	32.5	30.9	30.7	30.7	30.7	188.2	
Automatic debt dynamics [5/]	-0.5	0.3	1.1	2.7	2.6	1.7	0.9	0.9	1.0	9.9	
Interest rate/growth differential [6/]	-0.5	0.3	1.1	2.7	2.6	1.7	0.9	0.9	1.0	9.9	
Of which: real interest rate	0.1	1.3	1.7	2.5	2.7	2.5	2.1	2.0	2.0	13.9	
Of which: real GDP growth	-0.6	-1.0	-0.6	0.2	-0.1	-0.8	-1.2	-1.1	-1.0	-4.0	
Exchange rate depreciation [7/]	0.0	0.0	0.0	...	...	...	...	...	...	...	
Other identified debt-creating flows	-0.1	-1.5	0.9	1.6	1.3	0.8	0.0	0.0	0.0	3.7	
Deposits Drawdown (negative)	-0.1	-1.5	0.9	1.6	1.3	0.8	0.0	0.0	0.0	3.7	
Contingent liabilities	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Residual, including asset changes [8/]	-0.1	0.3	0.8	0.6	0.2	0.0	0.0	0.0	0.0	0.9	



Source: IMF staff calculations.

1/ Public sector is defined as non-financial public sector.

2/ Based on available data.

3/ Long-term bond spread over U.S. bonds.

4/ Defined as interest payments divided by debt stock (excluding guarantees) at the end of previous year.

5/ Derived as [(r - π(1+g) - g + ae(1+r))/(1+g+π+gπ)) times previous period debt ratio, with r = interest rate; π = growth rate of GDP deflator; g = real GDP growth rate; a = share of foreign-currency denominated debt; and e = nominal exchange rate depreciation (measured by increase in local currency value of U.S. dollar).

6/ The real interest rate contribution is derived from the numerator in footnote 5 as r - π (1+g) and the real growth contribution as -g.

7/ The exchange rate contribution is derived from the numerator in footnote 5 as ae(1+r).

8/ Includes asset changes and interest revenues (if any). For projections, includes exchange rate changes during the projection period.

9/ Assumes that key variables (real GDP growth, real interest rate, and other identified debt-creating flows) remain at the level of the last projection year.

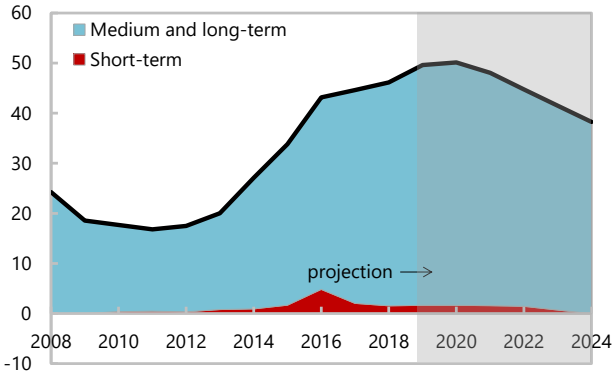
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Ecuador Public DSA - Composition of Public Debt and Alternative Scenarios

Composition of Public Debt

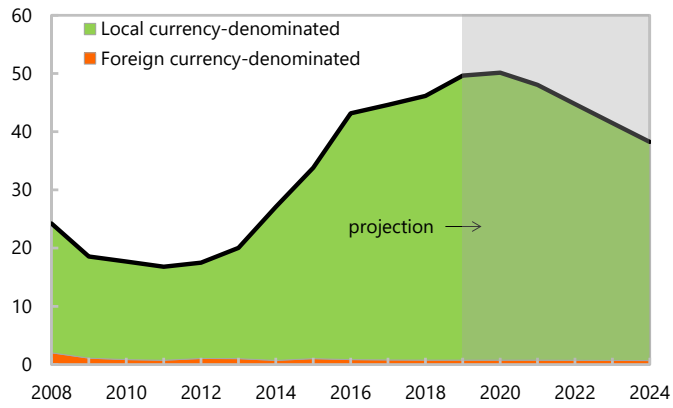
By Maturity

(in percent of GDP)



By Currency

(in percent of GDP)

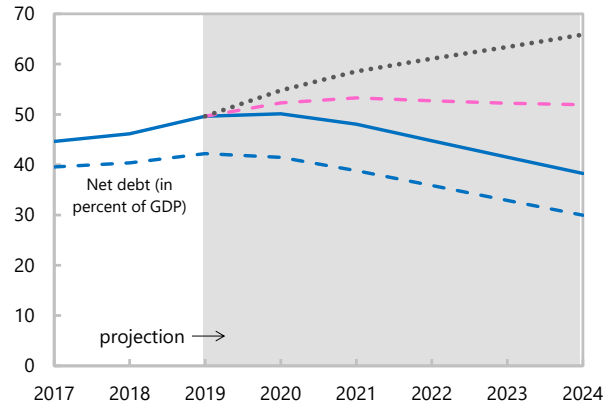


Alternative Scenarios

Baseline — Historical — Constant Primary Balance

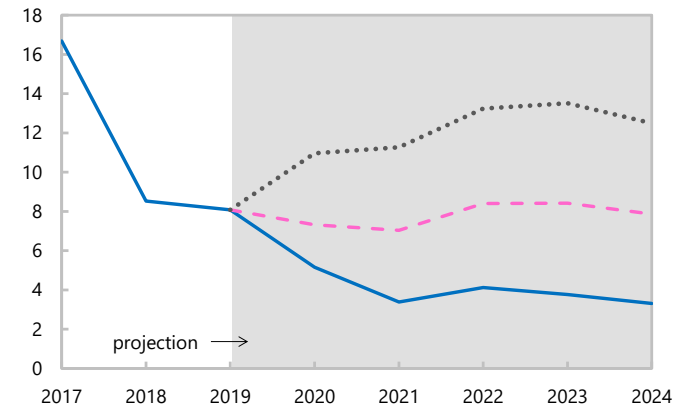
Gross Nominal Public Debt

(in percent of GDP)



Public Gross Financing Needs

(in percent of GDP)



Underlying Assumptions

(in percent)

Baseline Scenario	2019	2020	2021	2022	2023	2024
Real GDP growth	-0.5	0.2	1.6	2.7	2.5	2.5
Inflation	0.2	0.5	0.7	1.0	1.1	1.2
Primary Balance	1.5	3.7	4.6	4.2	4.2	4.3
Effective interest rate	5.6	6.0	5.9	5.7	5.7	6.3
Constant Primary Balance Scenario						
Real GDP growth	-0.5	0.2	1.6	2.7	2.5	2.5
Inflation	0.2	0.5	0.7	1.0	1.1	1.2
Primary Balance	1.5	1.5	1.5	1.5	1.5	1.5
Effective interest rate	5.6	6.0	5.8	5.5	5.6	6.1

Historical Scenario	2019	2020	2021	2022	2023	2024
Real GDP growth	-0.5	2.9	2.9	2.9	2.9	2.9
Inflation	0.2	0.5	0.7	1.0	1.1	1.2
Primary Balance	1.5	-2.4	-2.4	-2.4	-2.4	-2.4
Effective interest rate	5.6	6.0	5.0	4.2	4.0	4.3

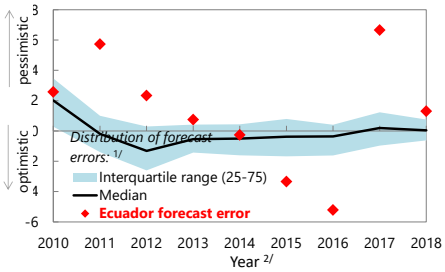
Source: IMF staff calculations.

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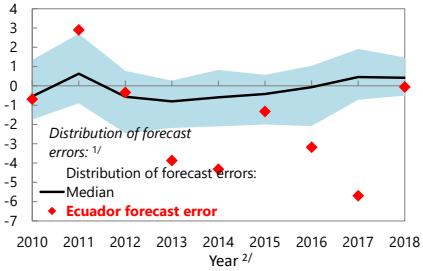
Ecuador Public DSA - Realism of Baseline Assumptions

Forecast Track Record, versus all countries

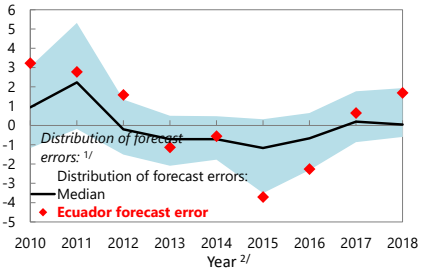
Real GDP Growth
(in percent, actual-projection)
Ecuador median forecast error, 2010-2018: 1.31
Has a percentile rank of: 92%



Primary Balance
(in percent of GDP, actual-projection)
Ecuador median forecast error, 2010-2018: -1.33
Has a percentile rank of: 17%

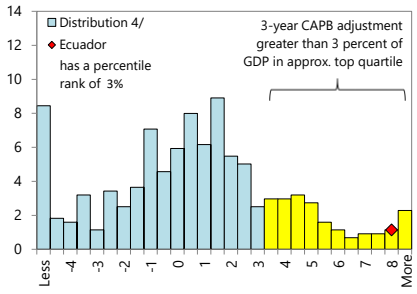


Inflation (Deflator)
(in percent, actual-projection)
Ecuador median forecast error, 2010-2018: 0.64
Has a percentile rank of: 71%

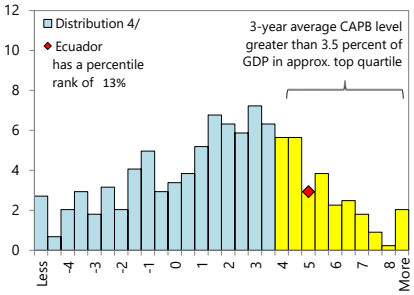


Assessing the Realism of Projected Fiscal Adjustment

3-Year Adjustment in Cyclically-Adjusted Primary Balance (CAPB)
(Percent of GDP)

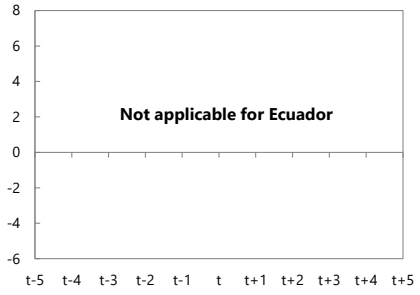


3-Year Average Level of Cyclically-Adjusted Primary Balance (CAPB)
(Percent of GDP)



Boom-Bust Analysis 3/

Real GDP growth
(in percent)



Source : IMF Staff calcuulations.
1/ Plotted distribution includes all countries, percentile rank refers to all countries.
2/ Projections made in the spring WEO vintage of the preceding year.
3/ Not applicable for Ecuador, as it meets neither the positive output gap criterion nor the private credit growth criterion.
4/ Data cover annual obervations from 1990 to 2011 for advanced and emerging economies with debt greater than 60 percent of GDP. Percent of sample on vertical axis.

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